# AN ANALYSIS AND SIMULATION OF DEFENDING TRACKING CONTROL NEURAL NETWORK FOR KNOWN AFFINE DISCRETE-TIME NONLINEAR SYSTEMS AGAINST ADVERSARIAL ATTACKS

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## Abstract

A neural network (NN) tracking control for discrete-time nonlinear systems with known internal dynamics is designed. The adversarial attack on the tracking control NN is modeled as additive stochastic noise injected into the output of the signal function of the controller. The adversarially perturbed system and the corresponding closed-loop tracking dynamics are derived, and the resulting degradation in tracking performance is demonstrated. The defense is based on rounding the noisy output of the signal function. When the perturbation magnitude is less than one half, the rounding recovers the correct signal function value, and the defended closed-loop dynamics reduce to the unperturbed system. The simulation is implemented in both MATLAB and C++; element-wise agreement of the state trajectories to machine precision is obtained, and the elapsed time of the C++ implementation is less than that of the MATLAB implementation by approximately two orders of magnitude.

## Introduction

The tracking control problem is one of the most fundamental practical problems in control theory, where the goal is to design a controller that allows a system to follow a desired trajectory or reference signal over time. In recent years, neural networks have been increasingly used in control systems due to their ability to approximate complex nonlinear functions and adapt to changing conditions. However, the use of neural networks in control systems also introduces new vulnerabilities, particularly to adversarial attacks. Small, practically undetectable perturbations in the input of a neural network can lead to outsized adverse effects on the output, which can be particularly problematic in control systems where safety and reliability are paramount.

A tracking control neural network (NN) for the known affine in the inputs, discrete-time nonlinear system was presented in [1]. The paper describes a method of tracking control that is of moderate complexity and requires no training. This method uses a signum activation function of the tracking error, as well as a diagonal matrix of the learning rate parameters to drive the system state towards the reference trajectory in finite time. The method is inherently robust to bounded uncertainties and disturbances.

## Mathematical Model and Block Diagram

The known affine in the inputs, discrete-time nonlinear system obtained by Euler approximation can be presented as follows [1]:

$$x(k+1) = x(k) + \tau\big[f(x(k)) + g(x(k))u(k)\big] \quad (1)$$

where $x(k)$ is the discrete-time state, $f$ and $g$ are the internal and coupling dynamics, $u(k)$ is the control input, $\tau$ is the sampling step, and $e(k) = x(k) - r(k)$ is the tracking error with respect to the reference $r(k)$. The control input is given by [1]

$$u(k) = g(x(k))^{-1}\big[-f(x(k)) + \Delta r(k) - \beta\ \mathrm{sgn}(e(k))\big] \quad (2)$$

where $\beta = \mathrm{diag}(\beta_1, \ldots, \beta_n)$ is the diagonal matrix of learning-rate parameters and $\Delta r(k) = r(k+1) - r(k)$ is the first-order finite difference of the reference. Substituting (2) into (1) yields the tracking error recursion

$$e(k+1) = e(k) - \beta\ \mathrm{sgn}(e(k)) \quad (3)$$

The adversarial attack is modeled [3] as an additive stochastic perturbation $\epsilon(k)$ with normal distribution and zero mean, injected at the output of the signal function:

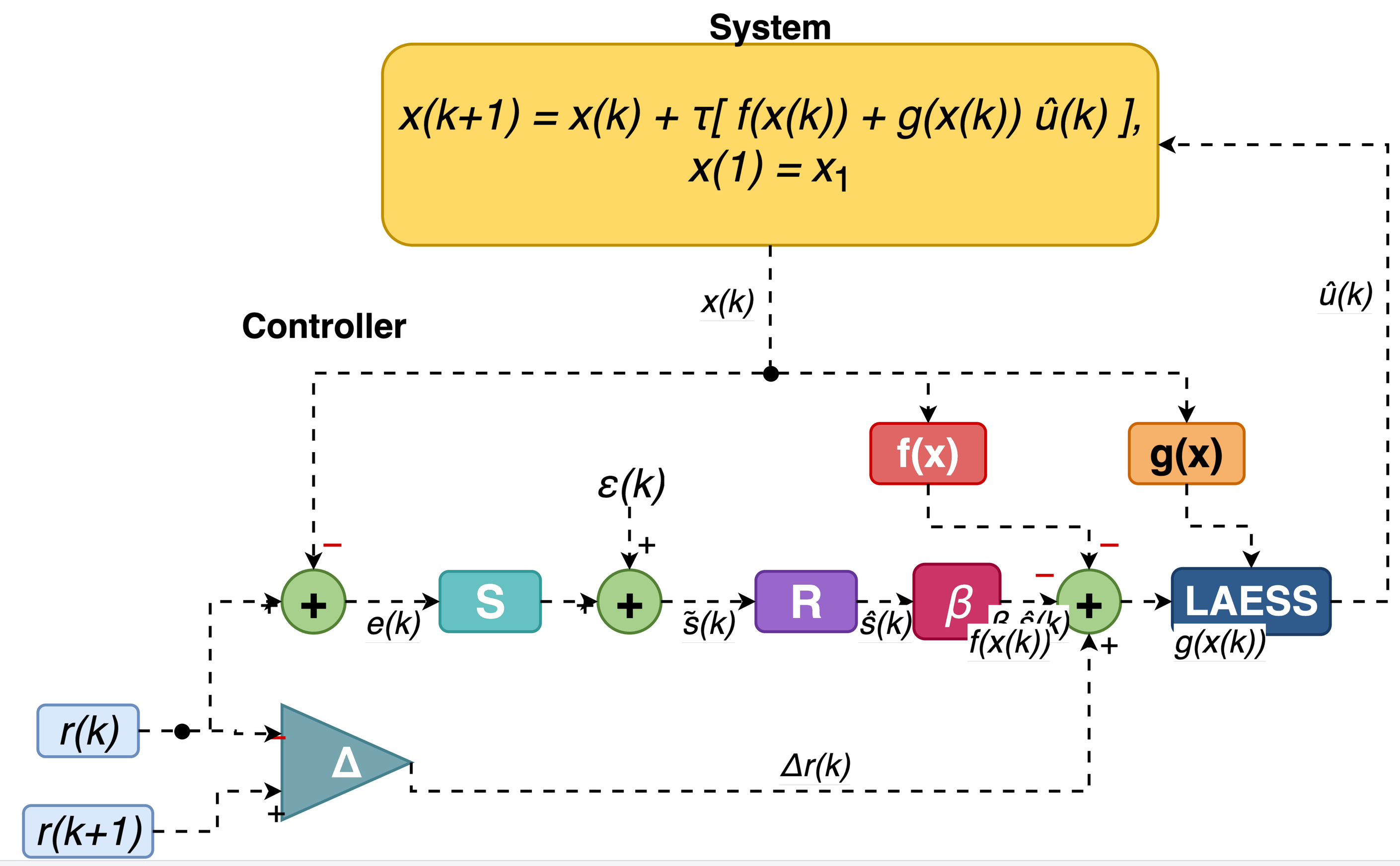
$$\tilde{s}(k) = \mathrm{sgn}(e(k)) + \epsilon(k) \quad (4)$$

Since $\mathrm{sgn}(e(k)) \in \{-1, 0, +1\}$, the defended output obtained by nearest-integer rounding

$$\hat{s}(k) = \mathrm{round}(\tilde{s}(k)) \quad (5)$$

satisfies $\hat{s}(k) = \mathrm{sgn}(e(k))$ whenever $|\epsilon(k)| < \frac{1}{2}$, so the defended closed-loop state equation reduces to the unperturbed system.

The defended closed-loop functional block diagram derived from equations (1)–(5) is shown in Fig. 1.




## Functional Description and Comparative Analysis

The controller of Fig. 1 implements supervised learning of the state $x(k)$ towards the reference $r(k)$ with learning rate $\beta$, which determines the convergence rate of $e(k)$ to zero. The signum activation function and the diagonal learning-rate matrix drive the tracking error to zero in finite time and impart inherent robustness to bounded uncertainties and disturbances [1].

Compared with the neural network optimal control approach for discrete-time nonlinear systems with known internal dynamics in [2], which minimizes a quadratic cost functional and requires offline training, the controller of Fig. 1 is of moderate complexity and requires no training. The block diagram differs from the unperturbed analog of [1] by the addition of the summation block injecting $\epsilon(k)$ at the output of S and the rounding block R that defends the control law against the additive perturbation. Compared with the projected gradient descent defense of [4], which is formulated for static classification networks, the rounding defense exploits the discrete codomain $\{-1, 0, +1\}$ of the signum activation and admits a closed-form recovery condition $|\epsilon(k)| < \frac{1}{2}$.

The elapsed time of the full simulation in MATLAB and C++ is compared in Table 1: the C++ implementation runs by approximately two orders of magnitude faster.

**Table 1.** Elapsed time of the full CSTR simulation ($n = 45{,}000$ steps, $\tau = 0.001$ s).

| Configuration | MATLAB (s) | C++ (s) |
|---|---|---|
| Unperturbed | $6.92 \times 10^{-1}$ | $1.29 \times 10^{-3}$ |
| Attacked | $6.46 \times 10^{-1}$ | $1.35 \times 10^{-3}$ |

## Simulation Results

The closed-loop dynamics are simulated on the continuous stirred-tank reactor (CSTR) benchmark with $\alpha = 1.0$, $\lambda = 0.3$, $\gamma = 20.0$, $B = 1.0$, $D = 0.072$, $\beta = 100$, and sampling step $\tau = 0.001$ s, over $n = 45{,}000$ iterations. At each time step a Gaussian noise with zero mean and standard deviation $\sigma = 0.5$ is added to the output of the signal function as in (4) and rejection-sampled to satisfy $|\epsilon(k)| < \frac{1}{2}$, so the perturbation is bounded by the rounding-recovery threshold in every realization. Three configurations—unperturbed, attacked, and defended—are simulated in both MATLAB and C++.

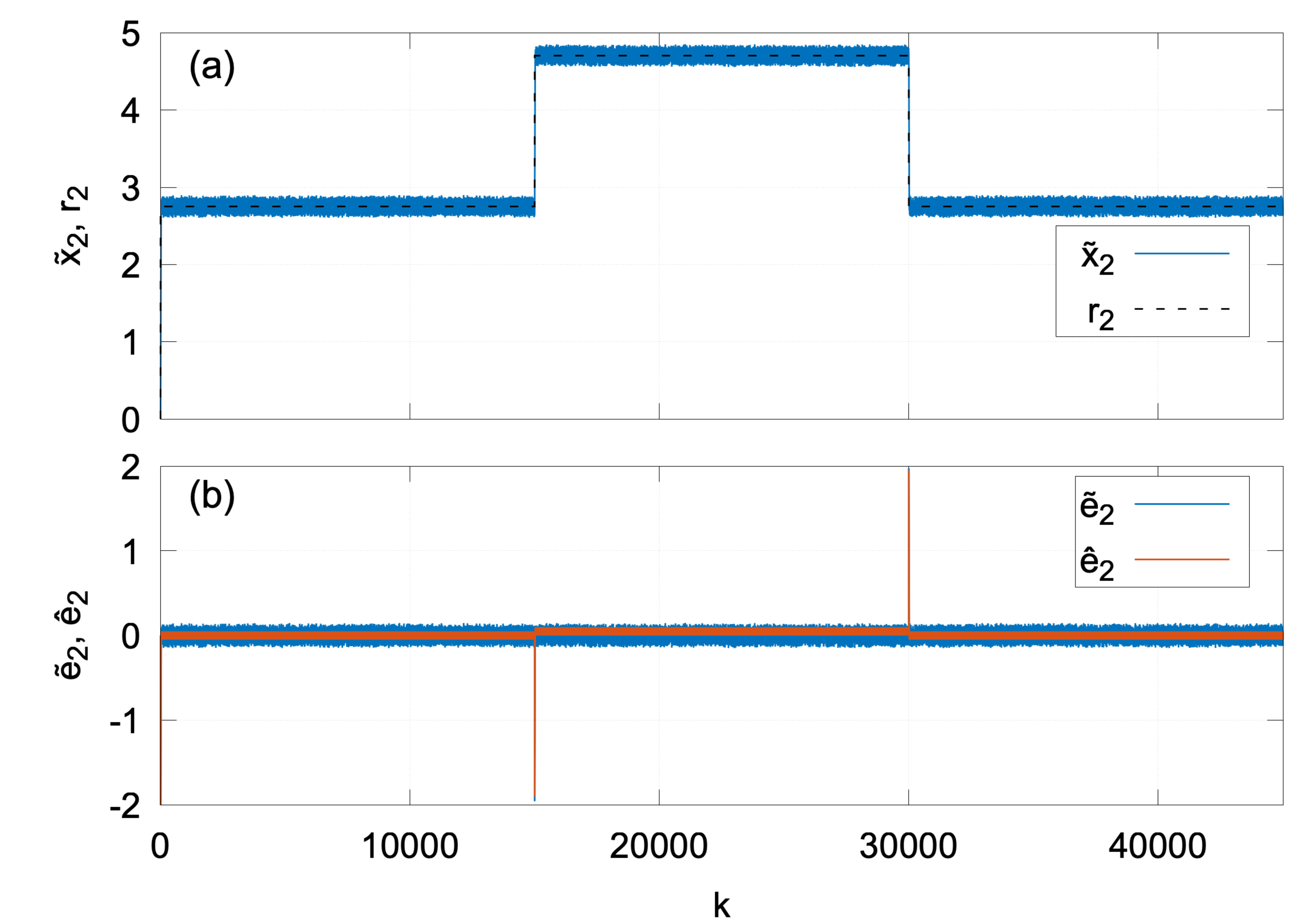



**Fig. 2.** CSTR closed-loop response under additive stochastic attack, with the tilde marking perturbed quantities and the hat marking defended quantities: **(a)** perturbed state variable $\tilde{x}_2$ (solid blue) against reference $r_2$ (dashed black); **(b)** perturbed tracking error $\tilde{e}_2(k) = \tilde{x}_2(k) - r_2(k)$ (blue) and defended tracking error $\hat{e}_2(k) = \hat{x}_2(k) - r_2(k)$ (orange) over the full trajectory of $n = 45{,}000$ iterations.

As shown in Fig. 2(a), the attacked CSTR exhibits degraded tracking with visible oscillations of $\tilde{x}_2$ around the reference $r_2$. In Fig. 2(b), the defended tracking error $\hat{e}_2$ coincides with that of the unperturbed system at every step, while the perturbed tracking error $\tilde{e}_2$ remains nonzero. The state and tracking error trajectories are element-wise identical between the MATLAB and C++ implementations to machine precision; the corresponding elapsed-time comparison is reported in Table 1 (column 2).

## Conclusions

A defense based on rounding the noisy output of the signal function of the tracking control neural network is formulated and analyzed for known affine discrete-time nonlinear systems. The two main results are: (i) when the perturbation magnitude satisfies $|\epsilon(k)| < \frac{1}{2}$, the rounding defense recovers the clean signal value and the defended closed-loop dynamics coincide with the unperturbed system; and (ii) the C++ implementation reproduces the MATLAB reference results with element-wise agreement to machine precision while reducing the elapsed time of the full simulation by approximately two orders of magnitude, enabling embedded deployment without numerical degradation.

## References

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## Future Work

Future work extends the threat model from additive stochastic noise to projected gradient descent (PGD) adversarial attacks [4], analyzes attacks injected directly on the state input $x(k)$ and the reference trajectory $r(k)$ rather than at the output of the signal function, generalizes the rounding defense to quantization-aware schemes for continuous activations, and carries out hardware-in-the-loop validation of the C++ controller on FPGA and Arduino targets, including piezoelectric positioning stages.

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